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Technical Information Document

# DXT Data Management System Data Interface for ALYX

The purpose of this document is to define the interface specification required for communication between DXT and external Data Management System (DMS).

This specification defines the physical interface, data communication protocol, and data content of messages passed between DXT and the DMS.

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# Chapter 1 General Description

DXT is built using REST principles which ensure predictable URLs that make writing interface easy. The DXT interfaces follow HTTP rules, enabling a wide range of HTTP clients and servers to be used to interact with DXT.

This chapter gives an overview of DXT interface structure.

- Overview
- Web Services
  - Procedure Parameters
  - Procedure Parameters Status
  - Electronic Record
- Message Format
  - HTTP Request
  - HTTP Response
  - Root Element

## Section 1.1 Overview

DXT implements bi–directional communications with external data management systems (DMS) to provide asynchronous communications. Based on the type of the message, the communications can be initiated by either DXT or the DMS.

For example, when the DMS sends procedure setup file to DXT, the DMS will act as a client posting setup data to the DXT server. Likewise, when DXT sends an electronic record to the DMS, DXT will act as a client posting procedure data to the DMS server.

## Section 1.2 Web Services

This section provides an overview of the web services included with DXT. The following chapters contain detailed description of each service as well as structure and content of each message.

### Procedure Parameters

The DMS will use the Procedure Parameters service to send a procedure setup file to DXT.


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The procedure setup parameters include details about the donor to enable the instrument to determine the appropriate procedure settings for the donor.

The main purpose of remote procedure setup is to help improve procedure setup time and help reduce manual entry errors at the instrument. DXT uses pull method to achieve this.

**Pull Method**

In the pull model of remote procedure setup, the DMS provides the procedure setup file to DXT. DXT will queue the information, making it visible in DXT in the Monitoring feature, on the Donation Setup tab. When an operator enters or scans the Donation ID or Donation Setup ID into the apheresis instrument, the instrument triggers a request for the procedure seup file from DXT. This is commonly referred to as the pull model since the instrument is pulling the data from DXT.

The following sequence diagram depicts interactions between DMS, DXT, instrument, and operator for the pull model.

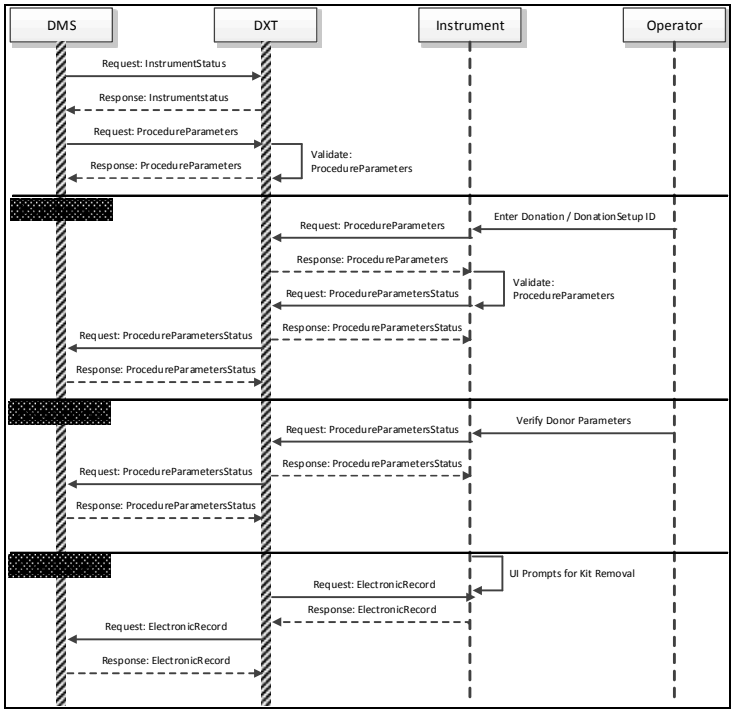



**Figure 1 Typical Sequence of Interactions in PULL Mode**

### Procedure Parameter Status

DXT will use the Procedure Parameter Status service to notify the DMS of the status of the procedure parameter.

### Electronic Record

DXT will use the Electronic Record service to provide the procedure record data to the DMS.


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DXT receives and stores data from connected instruments. Using the Electronic Record service, DXT can forward the data to the DMS. The data maintained in DXT can be considered as the 21 CFR Part 11 and/or EU Annex 11 compliant electronic records for a blood or platelet center, provided the center has policies and procedures for system usage as described in the regulation. Refer to DXT Data Management System Administrator's Guide Chapter 7 – Cyber Security, Part 11, and Annex 11 for considerations for deploying DXT in a 21 CFR Part 11 and/or EU Annex 11 compliant environment.

#### Instrument Status

The DMS will use the Instrument Status service to determine the availability of the site's instruments for remote procedure setup.

## Section 1.3 Message Format

#### HTTP Request

When DXT acts as a client, it will send all HTTP request headers in the following format. When DXT acts as a server, it expects all HTTP request headers in the same format.

```
METHOD URL HTTP/1.1 Host: SERVER_ADDRESS
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Connection: Keep-Alive
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

Method must be either GET or POST, DXT does not currently support resources updates (PUT) or deletion (DELETE). DXT will return an error response to HTTP PUT and HTTP DELETE requests.

HTTP header **Host** must be present and contain network name of the destination server.

HTTP Header **Content–Length** must be present and contain payload (XML) length in bytes.

HTTP Header **Content–Type** must be "text/xml; charset=utf–8", unless otherwise noted for the specific message.

HTTP Header **Date** must be present and contain date and time when message was generated using full Date format, specified by RFC–2616, Hypertext Transfer Protocol –– HTTP/1.1. For example:

```
Date: Fri, 14 Jun 2013 21:12:59 GMT
```

DXT expects all messages to be received within 2 minutes to ensure proper functionality. DXT will respond with an error to any message that contains an HTTP header Date Time more than 2 minutes in the past or in the future from DXT server time.

HTTP Header must include **Connection: Keep–Alive** directive.

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HTTP Header must include **Cache–Control** with cache directive set to “no–cache” and “no–transform”. Comma “,” and additional values (e.g., “private”) are acceptable.

The value of HTTP Header **FK–SourceId** and **FK–DestinationId** must be set as following:

For DXT, it should be “0”.
For DMS, it should be “1”.
For QC System, it should be “2”.

DXT expects **FK–SourceId** in HTTP Header of all inbound messages, and will supply **FK–DestinationId** in HTTP Header of all outbound messages.

For example, when the DMS sends data to DXT as a server, the IDs will be as following:

```
FK-SourceId: 1
FK-DestinationId: 0
```

## Section 1.4 Root Element

All XML messages must have a root element of DMSMessage as shown below. No data is associated with this element, it is a container for all child elements.

```
<xsd:element name="DMSMessage">
    <xsd:complexType>
        <xsd:sequence>

            <CHILD ELEMENTS>

            </xsd:sequence>
</xsd:complexType>
</xsd:element>
```

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# Chapter 2 DXT Server

This chapter defines messages that DXT accepts from external data management systems.

For possible error responses from DXT refer to Error Handling section at the end of this chapter.

## Section 2.1 Procedure Parameters

The DMS will use the Procedure Parameters service to send procedure parameters to DXT. This enables the remote procedure setup if DXT is licensed for this feature.

### Procedure Parameters URL

```
POST /fresenius-kabi/dxt/dms/procedure-parameters HTTP/1.1
```

Procedure Parameters request to DXT must supply ProcedureParameters XML object.

### Procedure Parameters Content Type

Procedure Parameters request to DXT must contain the following content type header element:

```
Content-Type: multipart/mixed;
boundary=this_is_fresenius_kabi_http_boundary
```

The first part of the message must have content type “text/xml; charset=utf–8” and XML parameters file. Other parts are reserved for future use and should not be supplied:

```
--this_is_fresenius_kabi_http_boundary
Content-Type: text/xml; charset=utf-8

<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <ProcedureParameters>

        ...

    </ProcedureParameters>

</DMSMessage>
--this_is_fresenius_kabi_http_boundary--
```

### ProcedureParameters XSD Definition

This complex type defines the contents of the Procedure Parameters structure as shown below.

```
<xsd:complexType name="ProcedureParameters">
    <xsd:sequence>

        <xsd:element name="DonationSetupId" minOccurs="1" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
```


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```
                    <xsd:pattern value="[a-zA-Z0-9]*" />
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="SiteIdentifier" minOccurs="1" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="20" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="DXTProcedureParameters" type="xsd:anyType" />

    </xsd:sequence>
</xsd:complexType>
```

### ProcedureParameters Data Description

The following table explains the content of Procedure Parameters message:

| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| ProcedureParameters | DMSMessage | | Container. This tag is required. |
| DonationSetupId | ProcedureParameters | Alphanumeric string. 25 characters max. | The unique identifier for the procedure setup. It needs to match the DONATION_ID field in DXTProcedureParameters element. |
| SiteIdentifier | ProcedureParameters | String. 20 characters max.<br><br>Must be a site name defined in DMS configurations for the site. | The DMS Site Id associated with the Site Name configured in DXT. Refer to DXT Data Management System Administrator's Guide Chapter 4 for the information about DMS configurations.<br><br>SiteIdentifier is used to properly associate procedure parameters with the DXT site location for the displaying and reporting purposes. |
| DXTProcedureParameters | ProcedureParameters | | Procedure information complex element. The actual content depends on the instrument type as described in the following sections. |

User should verify that DMS Site ID on DXT matches Site ID programmed on the DMS. Incorrect DMS Site ID will prevent DXT from processing remote procedure parameters.

All procedure parameters delivery failures, except operator reject, procedure parameter XML validation failure, or invalid Site ID, are non–recoverable and such parameters must be resent from DMS with the new unique DonationSetupID.


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### ProcedureParameters DXT Response

Upon successful processing, DXT will respond to procedure parameters with the following XML message:

```
<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <ProcedureParameterStatus>
        <DonationSetupId>DONATION_SETUP_ID</DonationSetupId>
        <ProcedureParametersStatusCode>0</ProcedureParametersStatusCode>
    </ProcedureParameterStatus>
</DMSMessage>
```

DONATION_SETUP_ID in the response will always match the one provided in the request.

The status code in the initial response will always be 0 – Parameters are received by DXT. To monitor updates of Procedure Parameters Status updates, refer to Chapter 3.

## Section 2.2 Procedure Parameters for ALYX Instruments

The DMS will use the Procedure Parameters service to send procedure parameters to DXT. This enables the remote procedure setup in push mode if DXT is licensed for this feature.

Refer to Section 2.1 Procedure Parameters for overview of the message. This section defines content and provides examples for messages specific to ALYX instruments.

### DXTProcedureParameters XSD Definition

This complex type defines the contents of the Procedure Parameters structure as shown below.

```
<xsd:complexType name="DXTProcedureParameters">
    <xsd:sequence>
        <xsd:element name="Field" maxOccurs="unbounded">
            <xsd:complexType>
                <xsd:sequence>
                    <xsd:element name="Name" type="xsd:string" />
                    <xsd:element name="Value" type="xsd:string" />
                    <xsd:element name="DataType" type="xsd:string"
                                 minOccurs="0" />
                    <xsd:element name="Description" type="xsd:string"
                                 minOccurs="0" />
                    <xsd:element name="ConfirmationQuestion" type="xsd:string"
                                 minOccurs="0" />
                </xsd:sequence>
                <xsd:attribute name="FieldIndex" type="xsd:integer" />
            </xsd:complexType>
        </xsd:element>
        <xsd:element name="Questionnaire" minOccurs="0" maxOccurs="unbounded">
            <xsd:complexType>
                <xsd:sequence>
```


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```
                    <xsd:element name="Question" type="xsd:string" />
                    <xsd:element name="Answer" type="xsd:string" />
                </xsd:sequence>
                <xsd:attribute name="QuestionnaireIndex" type="xsd:integer" />
            </xsd:complexType>
        </xsd:element>
    </xsd:sequence>
    <xsd:attribute name="FieldCount" type="xsd:integer" />
    <xsd:attribute name="QuestionnaireCount" type="xsd:integer" />
</xsd:complexType>
```

### DXTProcedureParameters Data Description

The following table explains the content of Procedure Parameters message data for ALYX:

| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| DXTProcedureParameters | ProcedureParameters | | Container. This tag is required.<br>A “FieldCount” attribute identifies the number of Field elements in the container.<br>A “QuestionnaireCount” attribute identifies the number of Questionnaire elements in the container. |
| Field | DXTProcedureParameters | | Container, identifying each Field element in the DXTProcedureParameters. The element is always provided and can occur multiple times.<br>A “FieldIndex” attribute identifies the index of current Field. |
| Name | Field | String | Name of the field. This element is always provided.<br>Refer to the “ALYX Procedure Parameter Field Names” on the list of acceptable Field Name values accepted by ALYX instruments. |
| Value | Field | String | Value associated with Name field. This element is always provided. |
| DataType | Field | String | DataType associated with this field. |
| Description | Field | String | Description of the field. |
| ConfirmationQuestion | Field | String | Free text question optional to each field entry, requesting operator confirmation at the instrument. |

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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| Questionnaire | DXTProcedureParameters | | Container of free text questions to confirm the correct donor is at the instrument. These questions come after all Field entries.<br><br>This tag is required if using questionnaire.<br><br>A "QuestionnaireIndex" attribute identifies the index of current questionnaire. |
| Question | Questionnaire | String | Free text containing question and the answer candidate for operator to answer.<br><br>This field is required if questionnaire is present. |
| Answer | Questionnaire | String | An Yes/No answer to the question. Answer does not appear on the instrument screen. Input selected by operator must match the answer to confirm the procedure parameters. This field is required if using questionnaire.<br><br>The value must be either Yes or No. |

When the DMS includes free text questions with procedure parameters, donor's personally identifiable information and applicable laws need to be considered. Instrument will store such information at most for the duration of the procedure and DXT will store such information at most for 24 hours.

At least one confirmation or one free text question is required for the message to be accepted. It is acceptable to have multiple questions.

### ALYX Procedure Parameter Field Names

Following is the list of acceptable Field Name values on ALYX.

| Field Name | Data Type | Range | Description |
|---|---|---|---|
| DEVICE | String | ALYX | Required |
| DONATION_ID | String | Alphanumeric string. 25 characters max. | Required. It has to match DonationSetupId sent in ProcedureParameters. |
| DONOR_ID | String | Alphanumeric string. 25 characters max. | Optional |
| DONOR_WEIGHT | String | 50 – 453* (kg)<br>110 – 999* (lb) | Required |
| DONOR_WEIGHT_UNIT | String | kg<br>lb | Required |


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| Field Name | Data Type | Range | Description |
|---|---|---|---|
| DONOR_HEIGHT | Integer | 3 – 239* (cm)<br>1 – 94* (in) | Required |
| DONOR_HEIGHT_UNIT | String | cm<br>in | Required |
| DONOR_HCT | Integer | 38 – 55* | Required, if DONOR_GLOBIN is not present. |
| DONOR_GLOBIN | Float, with 1 digit after the decimal point | 12.5 – 18.3* | Required, if DONOR_HCT is not present. |
| DONOR_GENDER | String | M or F | Required |
| BLOOD_TYPE | String | A-, A+, B-, B+<br>AB-, AB+,<br>O-, O+, ? | Optional |
| DIA_PRESSURE | Integer | 50 – 110 | Optional |
| SYS_PRESSURE | Integer | 80 – 200 | Optional |
| DONOR_RATE | Integer | 30 – 120 | Optional |
| DONOR_TEMP | Float, with 1 digit after the decimal point | F 94 – 100;<br>C 34.4 – 37.8 | Optional |
| DONOR_TEMP_UNIT | String | C<br>F | Required, if DONOR_TEMP is present. |
| DONOR_ARM_USED | String | LEFT, RIGHT | Optional, identifying donor's arm to be used. |

*Note: (1) the ranges of acceptable donor parameters: donor height, weight, hemoglobin/hematocrit, are nomogram dependent and the combinations allowed may be rejected by the ALYX device. Please follow the nomograms found in the Operator's Manual.

**(2) The units of measure sent to the ALYX must match the unit of measure on the instrument, or the instrument will reject it.**

(3) Either DONOR_HCT or DONOR_GLOBIN is required. Both cannot be present in the message.

### Procedure Parameters Example for ALYX Instruments

Following is an example of complete Pull mode Procedure Parameters request for the ALYX instrument with software version 3.2 and 3.2B, where instrument is configured to use Donation ID as procedure setup file identifier.

```
POST /fresenius-kabi/dxt/dms/procedure-parameters HTTP/1.1
Date: Tue, 30 Oct 2018 16:00:19 GMT
Content-Type: multipart/mixed; boundary=this_is_fresenius_kabi_http_ boundary
Cache-Control: no-cache no-transform
FK-SourceId: 1
FK-Protocol-Version: 3.0.0.0
Host: 5.61.64.115
Content-Length: 1037
Expect: 100-continue
Connection: Keep-Alive
```


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```
--this_is_fresenius_kabi_http_boundary
Content-Type: text/xml; charset=utf-8

<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <ProcedureParameters>
        <DonationSetupId>001</DonationSetupId>
        <SiteIdentifier>DMS-Site1</SiteIdentifier>
        <DXTProcedureParameters FieldCount="12" QuestionnaireCount="2">
            <Field FieldIndex="1">
                <Name>DEVICE</Name>
                <Value>ALYX</Value>
            </Field>
            <Field FieldIndex="2">
                <Name>DONATION_ID</Name>
                <Value>001</Value>
                <ConfirmationQuestion>Please Confirm the donation ID is correct
             </ConfirmationQuestion>
            </Field>
            <Field FieldIndex="3">
                <Name>DONOR_ID</Name>
                <Value>286011</Value>
            </Field>
            <Field FieldIndex="4">
                <Name>DONOR_WEIGHT</Name>
                <Value>200</Value>
            </Field>
            <Field FieldIndex="5">
                <Name>DONOR_WEIGHT_UNIT</Name>
                <Value>lbs</Value>
            </Field>
            <Field FieldIndex="6">
                <Name>DONOR_HEIGHT</Name>
                <Value>77</Value>
            </Field>
            <Field FieldIndex="7">
                <Name>DONOR_HEIGHT_UNIT</Name>
                <Value>in</Value>
            </Field>
            <Field FieldIndex="8">
                <Name>DONOR_GLOBIN</Name>
                <Value>14.0</Value>
            </Field>
            <Field FieldIndex="9">
                <Name>DONOR_GENDER</Name>
                <Value>M</Value>
            </Field>
            <Field FieldIndex="10">
                <Name>BLOOD_TYPE</Name>
                <Value>O-</Value>
            </Field>
            <Field FieldIndex="11">
```



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```
            <Name>DONOR_TEMP</Name>
            <Value>98.6</Value>
        </Field>
        <Field FieldIndex="12">
            <Name>DONOR_TEMP_UNIT</Name>
            <Value>F</Value>
        </Field>
        <Questionnaire QuestionnaireIndex="1">
            <Question>What is your name?     BERNIE DAVID</Question>
            <Answer>Yes</Answer>
        </Questionnaire>
        <Questionnaire QuestionnaireIndex="2">
            <Question>What is the number of your street address?     1920
         </Question>
            <Answer>Yes</Answer>
        </Questionnaire>
    </DXTProcedureParameters>
</ProcedureParameters>
</DMSMessage>

--this_is_fresenius_kabi_http_boundary--
```

## Section 2.3 Instrument Status

The DMS will use the Instrument Status service to determine availability of the instruments for the remote procedure setup. As ALYX pulls the procedure parameters from DXT, this message is optional for DMS.

### Instrument Status URL

```
GET /fresenius-kabi/dxt/dms/instrument-status/{SiteId} HTTP/1.1
```

{SiteId} is the DMS Site Id associated with the Site Name configured in DXT. Refer to DXT Data Management System Administrator's Guide Chapter 4 for the information about DMS configurations.

### Instrument Status DXT Response

Upon successful processing, DXT will respond to instrument status request with the following XML Message:

```
<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <InstrumentStatus>
        ...
    </InstrumentStatus>
</DMSMessage>
```

### InstrumentStatus XSD Definition

This complex type defines the contents and structure of the Instrument Status DXT reply XML message.


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```
<xsd:complexType name="InstrumentStatus">
    <xsd:sequence>

        <xsd:element name="SiteIdentifier">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="20" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="Region">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="50" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="SiteName">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="50" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="Instrument" minOccurs="0" maxOccurs="unbounded">
            <xsd:complexType>
                <xsd:sequence>

                    <xsd:element name="InstrumentSerialNumber">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:minLength value="1" />
                                <xsd:maxLength value="20" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>

                    <xsd:element name="InstrumentName">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:minLength value="1" />
                                <xsd:maxLength value="20" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>
```



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```
                    <xsd:element name="InstrumentTypeCode" type="xsd:unsignedInt" />

                    <xsd:element name="CurrentInstrumentStatusCode"
                            type="xsd:unsignedInt" minOccurs="0" />

                    <xsd:element name="CurrentInstrumentStatusTime"
                            type="xsd:dateTime" minOccurs="0" />

                </xsd:sequence>
            </xsd:complexType>
        </xsd:element>
    </xsd:sequence>
</xsd:complexType>
```

### InstrumentStatus Data Description

The following table explains the content of the Instrument Status DXT reply message as applicable to ALYX instruments. Refer to the corresponding Data Management supplement for the description of instrument status data for other instruments.

| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| InstrumentStatus | DMSMessage | | Container. This element is always provided. |
| SiteIdentifier | InstrumentStatus | String. 20 characters max. | A DMS site name supplied in the request. |
| Region | InstrumentStatus | String. 50 characters max. | A name of the region configured in DXT. |
| SiteName | InstrumentStatus | String. 50 characters max. | A name of the site configured in DXT. |
| Instrument | InstrumentStatus | | Container. This element is repeated per instrument configured at the site. |
| InstrumentSerialNumber | Instrument | String. 20 characters max. | Serial Number of the instrument. This element is always provided. |
| InstrumentName | Instrument | String. 20 characters max. | Name of the instrument configured in DXT. This element is always provided. |
| InstrumentTypeCode | Instrument | Integer | Type of the apheresis instrument:<br><br>5 – Alyx<br><br>This element is always provided. |



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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| CurrentInstrumentStatusCode | Instrument | Integer | The last known instrument status:<br><br>1 – Available. Instrument ready to be setup.<br>2 – Reserved. Instrument setup has started, but collection has not started.<br>3 – Collecting. Collection has been started on the instrument.<br>5 – Disconnected. DXT has lost connection with the instrument.<br>9 – Misconfigured<br>11 – No Data Export<br><br>This element is omitted if there were no communications with the instrument since DXT Instance has started. |
| CurrentInstrumentStatusTime | Instrument | Integer | The UTC timestamp of the last status update received from the instrument (instrument status 1, 2, 3, 11).<br><br>The time when DXT lost communication with the instrument (instrument status 5), or when DXT determines the instrument is misconfigured (instrument status 9).<br><br>This element is omitted if there were no communications with the instrument since DXT Instance has started. |

If multiple DXT Instances are deployed, the Instrument Status request must be sent to the DXT instance hosting the site in the request. See DXT Data Management System Administrator's Guide Installation section for more information about using multiple DXT Instances.

### Instrument Status Response Example

Following is an example of Instrument Status DXT reply:

```
FK-Protocol-Version: 3.0.0.0
FK-DestinationId: 1
```


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```
Content-Length: 566
Cache-Control: no-cache, no-transform
Content-Type: text/xml; charset=utf-8
Date: Tue, 26 Mar 2019 16:03:39 GMT

<?xml version="1.0"?>
<DMSMessage>
    <InstrumentStatus>
        <SiteIdentifier>DMSSITE1</SiteIdentifier>
        <Region>DXTREG1</Region>
        <SiteName>SITE1</SiteName>
        <Instrument>
            <InstrumentSerialNumber>10001</InstrumentSerialNumber>
            <InstrumentName>ALYX001</InstrumentName>
            <InstrumentTypeCode>5</InstrumentTypeCode>
            <CurrentInstrumentStatusCode>5</CurrentInstrumentStatusCode>
            <CurrentInstrumentStatusTime>2019-03-26T16:03:14.512Z
          </CurrentInstrumentStatusTime>
        </InstrumentStatus>
</DMSMessage>
```

## Section 2.4 DXT Server Response

This section lists possible responses that DXT can return to the requests originated from the DMS.

### OK Response

DXT will return the following HTTP Response header when the request was processed successfully:

```
HTTP/1.1 200 OK
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Connection: Close
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

The response may or may not contain XML message, based on the request.

### Request Header Validation Error Response

When DXT encounters problems validating the request, the following response will be returned to the sender:

```
HTTP/1.1 400 Bad Request
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
```


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```
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>ERROR_CODE</Code>
        <Description>ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

The following table lists Error codes and descriptions for possible errors:

| Error Code | Error_Description | Notes |
|---|---|---|
| 51000 | FK–SourceId Null or Empty | Sender did not provide “FK–SourceId” header element in the request or the content of “FK–SourceId” header element is empty. |
| 51001 | FK–ProtocolVersion Null or Empty | Sender did not provide “FK–ProtocolVersion” header element in the request or the content of “FK–ProtocolVersion” header element is empty. |
| 51002 | Date Null or Empty | Sender did not provide “Date” header element in the request or the content of “Date” header element is empty. |
| 51003 | Content–Type Null or Empty | Sender did not provide “Content–Type” header element in the request or the content of “Content–Type” header element is empty. |
| 51004 | Host Null or Empty | Sender did not provide “Host” header element in the request or the content of “Host” header element is empty. |
| 51005 | Cache–Control Null or Empty | Sender did not provide “Cache–Control” header element in the request or the content of “Cache–Control” header element is empty. |
| 51006 | Bad FK–ProtocolVersion. Expected: expected version | FK–ProtocolVersion provided in the header is not compatible with DXT version installed. |
| 51007 | Inconsistent Content–Length Value, actual counted value: actual content length | Content–Length value provided in the header does not match the size of the message received. |
| 51008 | Unexpected Content– Type, Expected: Content– Type expected, Received: Content–Type received | Content–Type provided in the header does not match content–type that is expected for the request. |
| 51009 | XSD Validation Failure: xml error description | DXT Server has failed to validate the XML message body in the received request. Error description is provided in the “Description” element of the response XML. |
| 51010 | Invalid Cache–Control, Expected: Cache–Control expected | Cache–Control provided in the header does not contain “no–cache no–transform”. |

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### Request Time Validation Error Response

To ensure timely processing of all inbound messages, the DXT Server requires the message timestamp be within 2 minutes of the DXT Server time, when the message is received.

```
HTTP/1.1 403 Forbidden
Content–Length: 147
Content–Type: text/xml; charset=utf–8
Date: TIMESTAMP
Cache–Control: no–cache no–transform
FK–SourceId: 0
FK–DestinationId: DATA_DESTINATION
FK–Protocol–Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>51100</Code>
        <Description>Invalid Date value</Description>
    </ErrorResponse>
</DXTMessage>
```

### Request Length Validation Error Response

To ensure the integrity of all inbound messages, the DXT Server requires that each request contains Content–Length header element.

When request's Content–Length header element is missing or empty, the following response will be returned to the sender:

```
HTTP/1.1 411 Length Required
Content–Length: 157
Content–Type: text/xml; charset=utf–8
Date: TIMESTAMP
Cache–Control: no–cache no–transform
FK–SourceId: 0
FK–DestinationId: DATA_DESTINATION
FK–Protocol–Version: 3.0.0.0

 <DXTMessage>
     <ErrorResponse>
         <Code>51200</Code>
         <Description>Content–Length Null or Empty</Description>
         </ErrorResponse>
 </DXTMessage>
```

### Request HTTP Version Validation Error Response

When the DXT Server encounters unsupported HTTP version in the request, the following response will be returned to the sender:

```
HTTP/1.1 505 HTTP Version Not Supported
Date: TIMESTAMP
```


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```
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

This response does not contain XML message.

### Request Processing Error Response

When DXT encounters problems processing the request, the following response will be returned to the sender:

```
HTTP/1.1 500 Internal Server Error
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>ERROR_CODE</Code>
        <Description>ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

The following table lists Error codes and descriptions for possible errors:

| Error Code | Error Description | Notes |
|---|---|---|
| 50000 | Instrument Not Registered | The desired instrument is not registered with DXT or is registered through the DXT Instance other than one the request was sent to. |
| 50001 | Database Error | DXT has failed to access the database (read or write the data) to process the request. |
| 50009 | Remote Procedure Setup Not Licensed | DXT has received Procedure Parameters request, but Remote Procedure Setup feature is not licensed. |
| 50014 | Site ID Not Valid | SiteIdentifier element in Procedure Parameters request does not match Site ID configured in DXT for any site location. Or SiteId field in Instrument Status Request URL is not one of the Site Identifier entries configured in Site DMS configuration. |
| 50015 | Duplicate Donation Setup ID | DonationSetupId element in Procedure Parameters request contains duplicate / previously received identifier. |


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# Chapter 3 DXT Client

This chapter defines messages that DXT sends to external data management systems.

The destination URL is configurable through DXT User Interface for each data management system DXT communicates with. Refer to DXT Data Management System Administrator's Guide Chapter 4 – DXT Configurations on how to configure DMS interfaces.

For possible error responses from DXT refer to Error Handling section at the end of this chapter.

## Section 3.1 Electronic Records for ALYX Instruments

DXT will use the Electronic Record service to provide the procedure record data to the DMS for records received from ALYX instruments, if DXT is licensed for the electronic record feature and DMS is configured on DXT UI.

### Electronic Record URL

```
POST {Electronic Record URL}/procedure-record HTTP/1.1
```

DXT will send procedure records to the address configured as "Electronic Record URL" through DXT User Interface.

### ElectronicRecord XSD Definition

This complex type defines the contents of the Electronic Record structure. Due to the complexity, the definition is broken into several sub–types, each one presented separately.

```
<xsd:complexType name="ElectronicRecord"
                 xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <xsd:sequence>

        <xsd:element name="Procedure">
            <xsd:complexType>

                <xsd:all>

                    <xsd:element name="RecordNumber">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:maxLength value="50" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>

                    <xsd:element name="DonationID"
                            minOccurs="0">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:maxLength value="100" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>
```


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```
<xsd:element name="ProcedureType"
        minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="100" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="ProcedureStatus"
        minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="100" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="ProcedureLogStartTime"
        type="xsd:string"
        minOccurs="0" />

<xsd:element name="DriveLocation"
        minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="TeamLeader"
        minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="RecordType"
        type="xsd:integer">
</xsd:element>

<xsd:element name="CollectionStartTime"
        type="xsd:string"
        minOccurs="0" />

<xsd:element name="CollectionEndTime"
        type="xsd:string"
        minOccurs="0" />

<xsd:element name="CollectionDuration"
        minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
```



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```
            <xsd:extension base="xsd:string">
                <xsd:attribute name="units"
                      type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="NeedleInTime"
        type="xsd:string"
        minOccurs="0" />

<xsd:element name="NeedleOutTime"
        type="xsd:string"
        minOccurs="0" />

<xsd:element name="NINODuration"
        minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:string">
                <xsd:attribute name="units"
                      type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="PhlebotomistStartId"
        minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="100" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="PhlebotomistEndId"
        minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="100" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="KitPrimedId"
        minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="100" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="KitRemovedId"
        minOccurs="0">
```



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```
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="100" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

            </xsd:all>
        </xsd:complexType>
    </xsd:element>

    <xsd:element name="DeviceInfo" minOccurs="0">
        <xsd:complexType>
            <xsd:sequence>

                <xsd:element name="Device"
                        minOccurs="1">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="4" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

                <xsd:element name="SerialNumber"
                        minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="100" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

                <xsd:element name="InstrumentID"
                        minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="50" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

                <xsd:element name="SoftwareVersion"
                        minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="70" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

                <xsd:element name="InstrumentTimeZone"
                        minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="40" />
                        </xsd:restriction>
```



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```
                </xsd:simpleType>
            </xsd:element>

            <xsd:element name="Language"
                    minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="15" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>

        </xsd:sequence>
    </xsd:complexType>
</xsd:element>

<xsd:element name="Donor" minOccurs="0">
    <xsd:complexType>
        <xsd:all>

            <xsd:element name="DonorID"
                    minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="25" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>

            <xsd:element name="Gender" minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="6" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>

            <xsd:element name="Weight" minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                    type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Height" minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:string">
                            <xsd:attribute name="units"
                                    type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
```



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```
</xsd:element>

<xsd:element name="DonorArm" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="25" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="BloodType" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="3" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="Hematocrit" minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:integer">
                <xsd:attribute name="units"
                      type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="Hemoglobin" minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:float">
                <xsd:attribute name="units"
                      type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="BloodPressure" minOccurs="0">
    <xsd:complexType>
        <xsd:all>

            <xsd:element name="Diastolic">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                               type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Systolic">
                <xsd:complexType>
```



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```
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                           type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                    </xsd:all>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Temperature" minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:float">
                            <xsd:attribute name="units"
                                  type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="HeartRate" minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                  type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>

        </xsd:all>
    </xsd:complexType>
</xsd:element>

<xsd:element name="Targets">
    <xsd:complexType>
        <xsd:sequence>

            <xsd:element name="RBC" minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                  type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Plasma" minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
```



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```
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                  type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>

        </xsd:sequence>
    </xsd:complexType>
</xsd:element>

<xsd:element name="Consumables">
    <xsd:complexType>

        <xsd:choice maxOccurs="unbounded">

            <xsd:element name="Device">
                <xsd:complexType>
                    <xsd:all>

                        <xsd:element name="Name">
                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="50" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>

                        <xsd:element name="Code" minOccurs="0">
                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="100" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>

                        <xsd:element name="Lot" minOccurs="0">
                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="100" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>

                        <xsd:element name="Serial" minOccurs="0">
                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="100" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>

                        <xsd:element name="Expiration" type="xsd:string"
                             minOccurs="0" />

                        <xsd:element name="RawScan" minOccurs="0">
```



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```
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="100" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

            </xsd:all>
        </xsd:complexType>
    </xsd:element>

    <xsd:element name="Solution">
        <xsd:complexType>
            <xsd:all>

                <xsd:element name="Name">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="50" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

                <xsd:element name="Code" minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="100" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

                <xsd:element name="Lot" minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="100" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

                <xsd:element name="Expiration" type="xsd:string"
                     minOccurs="0" />

                <xsd:element name="RawScan" minOccurs="0">
                    <xsd:simpleType>
                        <xsd:restriction base="xsd:string">
                            <xsd:maxLength value="100" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>

            </xsd:all>
        </xsd:complexType>
    </xsd:element>

</xsd:choice>

<xsd:attribute name="count" type="xsd:integer" />
```



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```
    </xsd:complexType>
</xsd:element>

<xsd:element name="Results">
    <xsd:complexType>
        <xsd:all>

            <xsd:element name="WBProcessed" minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                                  type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="BloodLossVolumes" minOccurs="0">
                <xsd:complexType>
                    <xsd:sequence>

                        <xsd:element name="RBCsInKit" maxOccurs="unbounded">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                           type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                        <xsd:element name="PlasmaInKit" maxOccurs="unbounded">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                           type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                    </xsd:sequence>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="PercentageRecovery" type="xsd:integer"
                    minOccurs="0" />

            <xsd:element name="SalineUsed" minOccurs="0">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
```



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```
                    type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="ACUsed" minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:integer">
                <xsd:attribute name="units"
                    type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="RBCFiltration" minOccurs="0">
    <xsd:complexType>
        <xsd:sequence>

            <xsd:element name="Status" minOccurs="0">
                <xsd:simpleType>
                    <xsd:restriction base="xsd:string">
                        <xsd:maxLength value="150" />
                    </xsd:restriction>
                </xsd:simpleType>
            </xsd:element>

            <xsd:element name="Duration" type="xsd:string"
                 minOccurs="0" />

        </xsd:sequence>
    </xsd:complexType>
</xsd:element>

<xsd:element name="RBCTotalVolumeCollected" minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:integer">
                <xsd:attribute name="units"
                    type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="RBCAbsoluteVolumeCollected"
        minOccurs="0">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:integer">
                <xsd:attribute name="units"
                    type="xsd:string" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
```



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```
</xsd:element>

<xsd:element name="RBCProduct" minOccurs="0">
    <xsd:complexType>
        <xsd:all>

            <xsd:element name="Total">
                <xsd:complexType>
                    <xsd:sequence>

                        <xsd:element name="RBC">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                        type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                        <xsd:element name="Adsol">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                        type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                        <xsd:element name="AC">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                        type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                    </xsd:sequence>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Bag-A" minOccurs="0">
                <xsd:complexType>
                    <xsd:sequence>

                        <xsd:element name="RBC">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                        type="xsd:string" />
```



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```
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>

        <xsd:element name="Adsol">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                        type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>

        <xsd:element name="AC">
            <xsd:complexType>
                <xsd:simpleContent>
                    <xsd:extension base="xsd:integer">
                        <xsd:attribute name="units"
                        type="xsd:string" />
                    </xsd:extension>
                </xsd:simpleContent>
            </xsd:complexType>
        </xsd:element>

    </xsd:sequence>
</xsd:complexType>
</xsd:element>

<xsd:element name="Bag-B" minOccurs="0">
    <xsd:complexType>
        <xsd:sequence>

            <xsd:element name="RBC">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                            type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Adsol">
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:integer">
                            <xsd:attribute name="units"
                            type="xsd:string" />
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>
```



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```
                        <xsd:element name="AC">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                        type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                    </xsd:sequence>
                </xsd:complexType>
            </xsd:element>

        </xsd:all>
    </xsd:complexType>
</xsd:element>

<xsd:element name="PlasmaProduct" minOccurs="0">
    <xsd:complexType>
        <xsd:all>

            <xsd:element name="Total">
                <xsd:complexType>
                    <xsd:sequence>

                        <xsd:element name="Plasma">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                        type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                        <xsd:element name="AC">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:integer">
                                        <xsd:attribute name="units"
                                        type="xsd:string" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                    </xsd:sequence>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Bag-1" minOccurs="0">
                <xsd:complexType>
                    <xsd:sequence>
```



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```
                    <xsd:element name="Plasma">
                        <xsd:complexType>
                            <xsd:simpleContent>
                                <xsd:extension base="xsd:integer">
                                    <xsd:attribute name="units"
                                    type="xsd:string" />
                                </xsd:extension>
                            </xsd:simpleContent>
                        </xsd:complexType>
                    </xsd:element>

                    <xsd:element name="AC">
                        <xsd:complexType>
                            <xsd:simpleContent>
                                <xsd:extension base="xsd:integer">
                                    <xsd:attribute name="units"
                                    type="xsd:string" />
                                </xsd:extension>
                            </xsd:simpleContent>
                        </xsd:complexType>
                    </xsd:element>

                </xsd:sequence>
            </xsd:complexType>
        </xsd:element>

        <xsd:element name="Bag-2" minOccurs="0">
            <xsd:complexType>
                <xsd:sequence>

                    <xsd:element name="Plasma">
                        <xsd:complexType>
                            <xsd:simpleContent>
                                <xsd:extension base="xsd:integer">
                                    <xsd:attribute name="units"
                                    type="xsd:string" />
                                </xsd:extension>
                            </xsd:simpleContent>
                        </xsd:complexType>
                    </xsd:element>

                    <xsd:element name="AC">
                        <xsd:complexType>
                            <xsd:simpleContent>
                                <xsd:extension base="xsd:integer">
                                    <xsd:attribute name="units"
                                    type="xsd:string" />
                                </xsd:extension>
                            </xsd:simpleContent>
                        </xsd:complexType>
                    </xsd:element>

                </xsd:sequence>
            </xsd:complexType>
        </xsd:element>

    </xsd:all>
```



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```
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Samples" minOccurs="0">
                <xsd:complexType>
                    <xsd:sequence>

                        <xsd:element name="Sample"
                             maxOccurs="unbounded">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:string">
                                        <xsd:attribute name="Index"
                                           type="xsd:integer" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                    </xsd:sequence>
                    <xsd:attribute name="count"
                            type="xsd:integer" />

                </xsd:complexType>
            </xsd:element>

            <xsd:element name="ProductIDs" minOccurs="0">
                <xsd:complexType>
                    <xsd:sequence>

                        <xsd:element name="ProductID" maxOccurs="unbounded">
                            <xsd:complexType>
                                <xsd:simpleContent>
                                    <xsd:extension base="xsd:string">
                                        <xsd:attribute name="Index"
                                           type="xsd:integer" />
                                    </xsd:extension>
                                </xsd:simpleContent>
                            </xsd:complexType>
                        </xsd:element>

                    </xsd:sequence>
                    <xsd:attribute name="count" type="xsd:integer" />

                </xsd:complexType>
            </xsd:element>

        </xsd:all>
    </xsd:complexType>
</xsd:element>

<xsd:element name="ProductAlert" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:maxLength value="200" />
        </xsd:restriction>
    </xsd:simpleType>
```



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```
</xsd:element>

<xsd:element name="Events" minOccurs="0">
    <xsd:complexType>

        <xsd:choice maxOccurs="unbounded">
            <xsd:element name="Event">
                <xsd:complexType>
                    <xsd:sequence>

                        <xsd:element name="TimeStamp" type="xsd:string"
                             minOccurs="0" />
                        <xsd:element name="Description">

                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="100" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>

                        <xsd:element name="OperatorID" minOccurs="0">
                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="100" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>

                        <xsd:element name="EventCode" type="xsd:integer"
                             minOccurs="0" />

                    </xsd:sequence>
                </xsd:complexType>
            </xsd:element>

            <xsd:element name="Alarm">
                <xsd:complexType>
                    <xsd:sequence>

                        <xsd:element name="AlarmCode" type="xsd:integer" />

                        <xsd:element name="AlarmSeverity" type="xsd:integer" />

                        <xsd:element name="TimeStamp" type="xsd:string"
                             minOccurs="0" />

                        <xsd:element name="OperatorID" minOccurs="0">
                            <xsd:simpleType>
                                <xsd:restriction base="xsd:string">
                                    <xsd:maxLength value="100" />
                                </xsd:restriction>
                            </xsd:simpleType>
                        </xsd:element>

                        <xsd:element name="AlarmData1" type="xsd:integer" />
```



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```
                        <xsd:element name="AlarmData2" type="xsd:integer" />

                    </xsd:sequence>
                </xsd:complexType>
            </xsd:element>

        </xsd:choice>

        <xsd:attribute name="alarm-count" type="xsd:integer" />

        <xsd:attribute name="event-count" type="xsd:integer" />

    </xsd:complexType>
</xsd:element>
</xsd:sequence>

<xsd:attribute name="device" fixed="5" type="xsd:integer" />

</xsd:complexType>
```

### ElectronicRecord Data Description

The following table explains the content of ElectronicRecord:

| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| ProcedureRecord | DMSMessage | ElectronicRecord | Container. This element is always provided.<br><br>A “device” attribute identifies the device type:<br><br>5 – Alyx |
| Procedure | ProcedureRecord | | Container. This element is always provided. It contains data related to all types of procedure record generated by ALYX instrument. |
| RecordNumber | Procedure | String. 50 characters max. | A globally unique identifier of the procedure record. This element is always provided. |
| DonationId | Procedure | String. 100 characters max. | A unique identifier of a specific donation defined by the center. |
| ProcedureType | Procedure | String. 100 characters max. | The description of a procedure type performed on the instrument. |
| ProcedureStatus | Procedure | String. 100 characters max. | The final status of the complete or terminated procedure. |
| ProcedureLogStartTime | Procedure | String | The instrument local date and time when the procedure record was created. |
| DriveLocation | Procedure | String. 20 characters max. | The location entered by the operator recording where the blood drive is held. |



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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| TeamLeader | Procedure | String. 20 characters max. | The identifier for the supervisor of a team that manages the instrument during blood drives. |
| RecordType | Procedure | Integer | The unique identifier of a type of the procedure record generated on the apheresis instrument during its operation<br><br>1 – Donation<br>3 – Non–Donation<br>4 – Service records<br>5 – Demo/Simulation records |
| CollectionStartTime | Procedure | String | The Instrument local date and time when operator presses the donor draw start button on the instrument. |
| CollectionEndTime | Procedure | String | The Instrument local date and time when the instrument prompts the operator to disconnect the donor. |
| CollectionDuration | Procedure | String | The Collection duration. It is the elapsed time between CollectionStartTime and CollectionEndTime.<br><br>A “units” attribute identifies the duration unit. |
| NeedleInTime | Procedure | String | The Instrument local date and time when the operator confirmed of the needle stick. |
| NeedleOutTime | Procedure | String | The Instrument local date and time when the needle removal from the donor or patient is confirmed. |
| NINODuration | Procedure | String | The duration of the collection of procedure. It is the elapsed time between NeedleInTime and NeedleOutTime.<br><br>A “units” attribute identifies the duration unit. |
| PhlebotomistStartId | Procedure | String. 100 characters max. | The ID of the phlebotomist who performed the venipuncture on the donor. |
| PhlebotomistEndId | Procedure | String. 100 characters max. | The ID of the phlebotomist who removed the needle from the donor. |
| KitPrimedId | Procedure | String. 100 characters max. | The ID of the operator who initiated the apheresis procedure. |
| KitRemovedId | Procedure | String. 100 characters max. | The ID of the operator who confirmed the completion of the apheresis procedure. |


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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| DeviceInfo | ProcedureRecord | | Container |
| Device | DeviceInfo | String. 4 characters max. | The instrument type. Set to “ALYX” for ALYX instruments<br>This element is always provided. |
| SerialNumber | DeviceInfo | String. 100 characters max. | The instrument serial number that can be entered to the instrument manually by the site operator. |
| InstrumentID | DeviceInfo | String. 50 characters max. | Globally unique MAC (Media Access Control) address configured on the instrument. |
| SoftwareVersion | DeviceInfo | String. 70 characters max. | The software version. |
| InstrumentTimeZone | DeviceInfo | String. 40 characters max. | The time zone configured on the instrument. |
| Language | DeviceInfo | String. 15 characters max. | The language set on the instrument. |
| Donor | ProcedureRecord | | Container. The child elements can appear in any order. |
| DonorID | Donor | String. 25 characters max. | A unique identification number of the person who is donating at a specific instrument, as defined by the center. |
| Gender | Donor | String. 6 characters max. | A unique identifier of donor or patient gender. |
| Weight | Donor | Integer | The Donor’s weight measured before procedure.<br><br>A “units” attribute identifies the weight unit. |
| Height | Donor | String | The donor’s height measured before procedure.<br><br>A “units” attribute identifies the height unit. |
| DonorArm | Donor | String. 25 characters max. | The donor’s arm used for venipuncture. |
| BloodType | Donor | String. 3 characters max. | The Donor’s blood type.<br>Value example: “AB+”, “AB-”, “A+”, “A-”, “B+”, “B-”, “O+”, “O-”, “?”.<br>“?” is sent when blood type is unknown. |
| Hematocrit | Donor | Integer | The volume measure of the percentage of red blood cells in a donor’s blood before procedure.<br><br>A “units” attribute identifies the Hematocrit unit. |


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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| Hemoglobin | Donor | Float | The volume measure of the percentage of the oxygen carrying cells/proteins in a donor's blood before procedure.<br><br>A "units" attribute identifies the Hemoglobin unit. |
| BloodPressure | Donor | | Container.<br>The child elements can appear in any order. |
| Diastolic | BloodPressure | Integer | The donor's diastolic pressure in mm/Hg.<br><br>A "units" attribute identifies the Diastolic unit. |
| Systolic | BloodPressure | Integer | The donor's systolic pressure in mm/Hg.<br><br>A "units" attribute identifies the Temperature unit. |
| Temperature | Donor | Float | The donor's temperature in °C or °F.<br><br>A "units" attribute identifies the Diastolic unit. |
| HeartRate | Donor | Integer | The donor's heart rate in bpm.<br><br>A "units" attribute identifies the HeartRate unit. |
| Targets | ProcedureRecord | | Container |
| RBC | Targets | Integer | The target/desired amount of RBC to be collected.<br><br>A "units" attribute identifies the element unit. |
| Plasma | Targets | Integer | The target/desired amount of plasma to be collected.<br><br>A "units" attribute identifies the element unit. |
| Consumables | ProcedureRecord | | Container. It contains Disposable used during the procedure. The element can have multiple child elements. Child element is either Device or Solution.<br><br>A "count" attribute identifies the number of child elements. |



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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| Device | Consumables | | Container.<br><br>Its child elements can appear in any order. |
| Name | Device | String. 50 characters max. | The type of disposable.<br><br>This element is always provided for Device element. |
| Code | Device | String. 100 characters max. | The unique identifier associated with the consumable type.<br>Value:<br>1 – Device |
| Lot | Device | String. 100 characters max. | The Fresenius–Kabi assigned lot number to the manufacturing batch of the disposable. |
| Serial | Device | String. 100 characters max. | The Fresenius–Kabi assigned serial number to the manufacturing batch of the disposable. |
| Expiration | Device | String | The expiration date of the disposable used during the procedure. |
| RawScan | Device | String. 100 characters max. | The barcode information of the consumables. |
| Solution | | | Container.<br><br>Its child elements can appear in any order. |
| Name | Solution | String. 50 characters max. | The type of disposable.<br><br>This element is always provided for Device element. |
| Code | Solution | String. 100 characters max. | The unique identifier associated with the consumable type.<br>Value:<br>2 – Solution |
| Lot | Solution | String. 100 characters max. | The Fresenius–Kabi assigned lot number to the manufacturing batch of the disposable. |
| Expiration | Solution | String | The expiration date of the disposable used during the procedure. |
| RawScan | Solution | String. 100 characters max. | The barcode information of the consumables. |
| Results | ProcedureRecord | | Container.<br>Its child elements can appear in any order. |


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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| WBProcessed | Results | Integer | The Volume of whole blood processed.<br><br>A “units” attribute identifies the element unit. |
| BloodLossVolumes | Results | | Container.<br>It contains either RBCsInKit or PlasmaInKit element. |
| RBCsInKit | BloodLossVolumes | Integer | The absolute RBC Volume contained within kit.<br><br>A “units” attribute identifies the element unit. |
| PlasmaInKit | BloodLossVolumes | Integer | The plasma Volume contained within kit.<br><br>A “units” attribute identifies the element unit |
| PercentageRecovery | Results | Integer | The percentage of RBC product recovered in the red cell storage containers after filtration. |
| SalineUsed | Results | Integer | The actual volume of saline used or infused during procedure.<br><br>A “units” attribute identifies the element unit. |
| ACUsed | Results | Integer | The total volume of anticoagulant used during the procedure.<br><br>A “units” attribute identifies the element unit. |
| RBCFiltration | Results | | Container |
| Status | RBCFiltration | String. 150 characters max. | The filtration status from RBC product collection. |
| Duration | RBCFiltration | String | The total amount of time used to filter the red cells and preservation solution. |
| RBCTotalVolumeCollected | Results | Integer | Total volume of red cells, plasma, and anticoagulant in the red cell container.<br><br>A “units” attribute identifies the element unit |
| RBCAbsoluteVolumeCollected | Results | Integer | The volume of absolute (100% Hematocrit) red cells in the red cell container.<br><br>A “units” attribute identifies the element unit. |


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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| RBCProduct | Results | | Container.<br>Its child elements can appear in any order. |
| Total | RBCProduct | | Container |
| RBC | Total | Integer | The calculated volume of RBC product placed on the plasma scale for bag A and B.<br><br>A “units” attribute identifies the element unit. |
| Adsol | Total | Integer | The calculated volume of Adsol added to RBC Bag A and B.<br><br>A “units” attribute identifies the element unit. |
| AC | Total | Integer | The calculated volume of anticoagulant in RBC Bag A and B.<br><br>A “units” attribute identifies the element unit. |
| Bag–A | RBCProduct | | Container |
| RBC | Bag–A | Integer | The calculated volume of anticoagulant in RBC Bag A.<br><br>A “units” attribute identifies the element unit. |
| Adsol | Bag–A | Integer | The calculated volume of Adsol added to RBC Bag A.<br><br>A “units” attribute identifies the element unit. |
| AC | Bag–A | Integer | The calculated volume of anticoagulant in RBC Bag A.<br><br>A “units” attribute identifies the element unit. |
| Bag–B | RBCProduct | | Container |
| RBC | Bag–B | Integer | The calculated volume of anticoagulant in RBC Bag B.<br><br>A “units” attribute identifies the element unit. |
| Adsol | Bag–B | Integer | The calculated volume of Adsol added to RBC Bag B.<br><br>A “units” attribute identifies the element unit. |



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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| AC | Bag–B | Integer | The calculated volume of anticoagulant in RBC Bag B.<br><br>A “units” attribute identifies the element unit. |
| PlasmaProduct | Results | | Container.<br>Its child elements can appear in any order. |
| Total | PlasmaProduct | | Container |
| Plasma | Total | Integer | The calculated volume of plasma on the Plasma scale for bag 1 and 2.<br><br>A “units” attribute identifies the element unit. |
| AC | Total | Integer | The calculated volume of anticoagulant in the plasma product.<br><br>A “units” attribute identifies the element unit. |
| Bag–1 | PlasmaProduct | | Container |
| Plasma | Total | Integer | The calculated volume AC in the plasma on the Plasma scale for bag 1.<br><br>A “units” attribute identifies the element unit. |
| AC | Total | Integer | The calculated volume AC in the plasma on the Plasma scale for bag 1.<br><br>A “units” attribute identifies the element unit. |
| Bag–2 | PlasmaProduct | | Container |
| Plasma | Total | Integer | The calculated volume AC in the plasma on the Plasma scale for bag 2.<br><br>A “units” attribute identifies the element unit. |
| AC | Total | Integer | The calculated volume AC in the plasma on the Plasma scale for bag 2.<br><br>A “units” attribute identifies the element unit. |



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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| Samples | Results | | Container.<br><br>A “count” attribute identifies the number of child elements. |
| Sample | Samples | | The identifier associated with the 50 ml RBC sample packs.<br><br>This element is provided for each sample.<br><br>A “Index” attribute identifies the sample index. |
| ProductIDs | Results | | Container.<br><br>A “count” attribute identifies the number of child elements. |
| ProductID | ProductIDs | | The identifier associated with the product bag.<br><br>This element is provided for each product ID.<br><br>A “Index” attribute identifies the index of the associated product bag. |
| ProductAlert | Results | String. 200 characters max. | The text description of a product alert generated and reported by the instrument. |
| Events | ProcedureRecord | | Container.<br><br>A “alarm–count” attribute identifies the number of alarm elements<br><br>A “event–count” attribute identifies the number of event elements. |
| Event | Events | | Container.<br><br>The element is provided for each event. |
| TimeStamp | Event | String | The instrument local date and time when the event occurred. |
| Description | Event | String. 100 characters max. | The text description of an event. |
| OperatorID | Event | String. 100 characters max. | The ID of the operator signed into the instrument at the time of the event occurred. |



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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| EventCode | Event | Integer | The unique identifier associated with the event type.<br>21 – Alarm/Alert triggered by instruments.<br>22 – Instrument Events of Note (e.g. Kit Check Occurred) |
| Alarm | Events | | Container |
| AlarmCode | Alarm | Integer | The unique identifier associated with the alarm.<br><br>This element is always provided. |
| AlarmSeverity | Alarm | Integer | The unique identifier associated with the alarm severity.<br>0 – Warning1<br>1 – Warning2<br>2 – Cautionary0<br>3 – Cautionary1<br>4 – Cautionary2<br>5 – Cautionary3<br>6 – Internal<br><br>This element is always provided. |
| TimeStamp | Alarm | String | The instrument local date and time when the alarm occurred. |
| OperatorID | Alarm | String. 100 characters max. | The ID of the operator signed into the instrument at the time of the alarm occurred. |
| AlarmData1 | Alarm | Integer | The additional data provided by the instrument about an alarm event.<br><br>This element is always provided. |
| AlarmData2 | Alarm | Integer | The additional data provided by the instrument about an alarm event.<br><br>This element is always provided. |

DXT will forward donation records or non–donation records if the record type is selected during DMS configuration. Refer to DXT Data Management System Administrator's Guide Chapter 4 – DXT Configurations on how to configure DMS interfaces.

### Electronic Record Example for Donation Record

Following is an example of complete Procedure Record exchange between DXT and the DMS. The instrument type is ALYX with software version 3.2/3.2B. The RecordType is 1, i.e., Donation.


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```
POST /procedure-record HTTP/1.1
Connection: keep-alive,Keep-Alive
Date: Wed, 05 Jun 2019 15:58:07 GMT
FK-SourceId: 0
FK-Protocol-Version: 3.0.0.0
Cache-Control: no-transform, no-cache
Content-Type: text/xml; charset=utf-8
Host: 172.16.10.23:8060
Content-Length: 6776
```

```
<?xml version="1.0"?>
<DMSMessage>
    <ProcedureRecord>
        <ProcedureRecordAlyx device="5">
            <Procedure>
                <RecordNumber>00D0C9CFBBA9-PS-20180504-110139</RecordNumber>
                <DonationID>18109516H</DonationID>
                <ProcedureType>ALYX 2RBC-LR</ProcedureType>
                <ProcedureStatus>COMPLETE</ProcedureStatus>
                <ProcedureLogStartTime>5/4/2019 11:01:39 AM</ProcedureLogStartTime>
                <RecordType>1</RecordType>
                <CollectionStartTime>5/4/2019 11:07:11 AM</CollectionStartTime>
                <CollectionEndTime>5/4/2019 11:56:11 AM</CollectionEndTime>
                <NeedleInTime>5/4/2019 11:06:39 AM</NeedleInTime>
                <NeedleOutTime>5/4/2019 12:07:02 PM</NeedleOutTime>
                <NINODuration units="HH:MM:SS">00:00:00</NINODuration>
                <PhlebotomistStartId>DGK303</PhlebotomistStartId>
                <PhlebotomistEndId>DGK303</PhlebotomistEndId>
                <KitPrimedId>DGK303</KitPrimedId>
                <KitRemovedId>DGK303</KitRemovedId>
            </Procedure>
            <DeviceInfo>
                <SerialNumber>40768</SerialNumber>
                <SoftwareVersion>ALYX Release 3.2</SoftwareVersion>
                <InstrumentTimeZone>EST</InstrumentTimeZone>
                <Language>ENGLISH</Language>
            </DeviceInfo>
            <Donor>
                <Gender>Male</Gender>
                <Weight units="lb">175</Weight>
                <Height units="in">67</Height>
                <BloodPressure />
            </Donor>
            <Targets>
                <RBC units="ml">420</RBC>
            </Targets>
            <Consumables count="0" />
            <Results>
                <WBProcessed units="ml">1227</WBProcessed>
                <BloodLossVolumes>
                    <RBCsInKit units="ml">433</RBCsInKit>
                    <PlasmaInKit units="ml">107</PlasmaInKit>
```


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```
        </BloodLossVolumes>
        <PercentageRecovery>89</PercentageRecovery>
        <SalineUsed units="ml">434</SalineUsed>
        <ACUsed units="ml">142</ACUsed>
        <RBCFiltration>
            <Status>Complete Success</Status>
            <Duration>00:07:59</Duration>
        </RBCFiltration>
        <RBCTotalVolumeCollected units="ml">495</RBCTotalVolumeCollected>
        <RBCAbsoluteVolumeCollected units="ml">422
        </RBCAbsoluteVolumeCollected>
        <RBCProduct>
            <Total>
                <RBC units="ml">619</RBC>
                <Adsol units="ml">201</Adsol>
                <AC units="ml">100</AC>
            </Total>
            <Bag-A>
                <RBC units="ml">313</RBC>
                <Adsol units="ml">102</Adsol>
                <AC units="ml">313</AC>
            </Bag-A>
            <Bag-B>
                <RBC units="ml">307</RBC>
                <Adsol units="ml">100</Adsol>
                <AC units="ml">307</AC>
            </Bag-B>
        </RBCProduct>
        <PlasmaProduct>
            <Total />
            <Bag-1 />
            <Bag-2 />
        </PlasmaProduct>
        <Samples count="0" />
        <ProductIDs count="0" />
    </Results>
    <Events event-count="13" alarm-count="3">
        <Event>
            <TimeStamp>2019-05-04T10:53:15</TimeStamp>
            <Description>Operator Log In Occurred</Description>
            <OperatorID>DGK303</OperatorID>
            <EventCode>31</EventCode>
        </Event>
        <Event>
            <TimeStamp>2019-05-04T10:55:36</TimeStamp>
            <Description>Kit Check Occurred</Description>
            <OperatorID>DGK303</OperatorID>
            <EventCode>2</EventCode>
        </Event>
        <Event>
            <TimeStamp>2019-05-04T10:59:16</TimeStamp>
            <Description>Kit Prime Occurred</Description>
```




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```
    <OperatorID>DGK303</OperatorID>
    <EventCode>3</EventCode>
</Event>
<Event>
    <TimeStamp>2019-05-04T10:59:16</TimeStamp>
    <Description>Cannula Break Occurred</Description>
    <OperatorID>DGK303</OperatorID>
    <EventCode>32</EventCode>
</Event>
<Event>
    <TimeStamp>2019-05-04T10:59:56</TimeStamp>
    <Description>Saline Prime Occurred</Description>
    <OperatorID>DGK303</OperatorID>
    <EventCode>33</EventCode>
</Event>
<Event>
    <TimeStamp>2019-05-04T11:06:39</TimeStamp>
    <Description>Venipuncture Occurred</Description>
    <OperatorID>DGK303</OperatorID>
    <EventCode>22</EventCode>
</Event>
<Event>
    <TimeStamp>2019-05-04T11:06:40</TimeStamp>
    <Description>Start Collection Occurred</Description>
    <OperatorID>DGK303</OperatorID>
    <EventCode>23</EventCode>
</Event>
<Alarm>
    <AlarmCode>4001</AlarmCode>
    <AlarmSeverity>4</AlarmSeverity>
    <TimeStamp>2019-05-04T11:07:22</TimeStamp>
    <AlarmData1>13</AlarmData1>
    <AlarmData2>106</AlarmData2>
</Alarm>
<Event>
    <TimeStamp>2019-05-04T11:07:31</TimeStamp>
    <Description>Resume Collection Occurred</Description>
    <OperatorID>DGK303</OperatorID>
    <EventCode>26</EventCode>
</Event>
<Alarm>
    <AlarmCode>7026</AlarmCode>
    <AlarmSeverity>4</AlarmSeverity>
    <TimeStamp>2019-05-04T11:15:42</TimeStamp>
    <AlarmData1>50</AlarmData1>
    <AlarmData2>50</AlarmData2>
</Alarm>
<Event>
    <TimeStamp>2019-05-04T11:15:50</TimeStamp>
    <Description>Resume Collection Occurred</Description>
    <OperatorID>DGK303</OperatorID>
    <EventCode>26</EventCode>
```



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```
                </Event>
                <Alarm>
                    <AlarmCode>7026</AlarmCode>
                    <AlarmSeverity>4</AlarmSeverity>
                    <TimeStamp>2019-05-04T11:36:35</TimeStamp>
                    <AlarmData1>50</AlarmData1>
                    <AlarmData2>50</AlarmData2>
                </Alarm>
                <Event>
                    <TimeStamp>2019-05-04T11:36:41</TimeStamp>
                    <Description>Resume Collection Occurred</Description>
                    <OperatorID>DGK303</OperatorID>
                    <EventCode>26</EventCode>
                </Event>
                <Event>
                    <TimeStamp>2019-05-04T12:07:02</TimeStamp>
                    <Description>Disconnect Donor Occurred</Description>
                    <OperatorID>DGK303</OperatorID>
                    <EventCode>27</EventCode>
                </Event>
                <Event>
                    <TimeStamp>2019-05-04T12:09:35</TimeStamp>
                    <Description>Weigh Double RBC Product Occurred</Description>
                    <OperatorID>DGK303</OperatorID>
                    <EventCode>29</EventCode>
                </Event>
                <Event>
                    <TimeStamp>2019-05-04T12:11:10</TimeStamp>
                    <Description>General Event Occurred</Description>
                    <OperatorID>DGK303</OperatorID>
                    <EventCode>35</EventCode>
                </Event>
            </Events>
        </ProcedureRecordAlyx>
    </ProcedureRecord>
</DMSMessage>
```

## Section 3.2 Procedure Parameter Status

DXT will use the Procedure Parameter Status service to notify DMS on the status of the procedure parameter, if DXT is licensed for the remote procedure setup.

### Procedure Parameter Status URL

```
POST {Procedure Parameters URL}/procedure-parameter-status HTTP/1.1
```

DXT will send procedure parameter statuses to the address configured as “Procedure Parameters URL” through DXT User Interface.


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### ProcedureParameterStatus XSD Definition

This complex type defines the contents of the Procedure Parameters Status structure as shown below.

```
<xsd:complexType name="ProcedureParameterStatus">
    <xsd:sequence>
        <xsd:element name="DonationSetupId">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="ProcedureParametersStatusCode" type="xsd:unsignedInt" />

        <xsd:element name="ValidateFailureDetail" type="xsd:string" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>
```

### ProcedureParameterStatus Data Description

The following table explains the content of ProcedureParameterStatus:

| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| ProcedureParameterStatus | DMSMessage | | Container. This tag is required. |
| DonationSetupId | ProcedureParameter Status | Alphanumeric string. 25 characters max. | Same as the DonationSetupId field, which was sent from DMS in ProcedureParameter message.<br><br>This element is always provided. |

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| Element | Parent Element | Data Type | Description |
|---|---|---|---|
| ProcedureParametersStatusCode | ProcedureParameterStatus | Unsigned integer | Parameters status:<br><br>1 – parameters were received by the instrument;<br><br>2 – parameters were accepted by the instrument;<br><br>3 – parameters were rejected by the instrument. More information may be provided in ValidateFailureDetail element;<br><br>4 – parameters were accepted and/or confirmed by the operator at the instrument;<br><br>5 – parameters were rejected by the operator at the instrument;<br><br>8 – instrument is not in a valid state to accept parameters or is not configured to accept parameters in the push mode.<br><br>If neither confirmation questions nor questionnaire were supplied in procedure setup file, the status will be set to 4 after successful validation by the instrument.<br><br>This element is always provided. |
| ValidateFailureDetail | ProcedureParameterStatus | String | Description of the reason for instrument rejection of the parameters.<br><br>This element is provided only with Status Code 3. |

### ProcedureParameterStatus Example

Following is an example of complete Procedure Parameter Status between DXT and the DMS

```
POST /procedure-parameter-status HTTP/1.1
Connection: keep-alive,Keep-Alive
Date: Tue, 30 Oct 2018 16:11:22 GMT
FK-SourceId: 0
FK-DestinationId: 1
FK-Protocol-Version: 3.0.0.0
```


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```
Cache-Control: no-transform, no-cache
Content-Type: text/xml; charset=utf-8
Host: 127.0.0.1:8060
Content-Length: 220

<DMSMessage>
    <ProcedureParameterStatus>
        <DonationSetupId>14EILB093031591</DonationSetupId>
        <ProcedureParametersStatusCode>2</ProcedureParametersStatusCode>
    </ProcedureParameterStatus>
</DMSMessage>
```

## Section 3.3 DMS Response

This section lists responses that DXT expects from the DMS.

### OK Response

Upon successful processing, the DMS should respond to with the following:

```
Header: HTTP 1.1 200 OK
Content-Length: 0 Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0
```

### Error Response

When the DMS encounters problems validating the request header, the response to DXT should be formatted as following:

```
HTTP/1.1 {HTTP_ERROR_CODE} {HTTP_ERROR}
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>DMS_ERROR_CODE</Code>
        <Description>DMS_ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

DXT will treat any non–OK response as an error response. DXT does not specifically act on the type of the error response.

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When DXT receives any error response to any message, it will retry sending the message up to three times using the delay programmed for each DMS on DXT User Interface between each retry.

When sending Procedure Record, after all retry attempts are exhausted and transmission is still unsuccessful, DXT will record DMS_ERROR_ CODE and DMS_ERROR_DESCRIPTION of last unsuccessful attempt and will display them along with the record on the undelivered records screen as a failure code. If XML message body is not included in DMS error response or correctly formatted ErrorResponse element is not present in XML message, DXT will record and display HTTP_ERROR_ CODE and HTTP_ERROR instead. Refer to DXT Data Management System User's Guide Chapter 4 – Monitoring for an additional information on handling undelivered records.

When sending Procedure Parameter Status, after all retry attempts are exhausted and transmission is still unsuccessful, DXT will abandon transmission and no further attempts to resend status message will be made.

***Over time, this information may be subject to change based upon modification in product manufacturing, blood banking, or critical practice. For additional questions please contact the Clinical Technical Hotline at 1–800–937–5060.***

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